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Sleep Scoring App: Visualizing and annotating rodent sleep recordings with synchronized electrophysiology, photometry, and behavior video

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Software

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Summary

Sleep Scoring App (`sleep_scoring`) is a local desktop application for sleep scoring rodent recordings: labeling each second as wakefulness (Wake), non-rapid-eye-movement (NREM) sleep, REM sleep, or a brief microarousal (MA), from brain electrical activity (EEG) and muscle activity (EMG). It brings EEG, EMG, an EEG spectrogram, an optional norepinephrine (NE) signal measured by fiber photometry, and behavior-video clips of a selected interval into one scoring workflow. Users can label intervals manually or start from predicted sleep scores, then revise individual seconds or whole bouts; predictions never overwrite the user's own labels. The default rule-based scoring model adapts to the user's labels on each recording, and completed scores are exported as bout tables with stage and transition statistics.

Statement of need

Sleep Scoring App was developed for researchers studying how brain state regulates cerebrospinal-fluid transport during sleep, in Project 2 of the BrainFlowZZZ program at the University of Rochester. That work relates sleep states to cortical NE dynamics, which oscillate during NREM sleep and shape sleep architecture (Kjaerby et al., 2022) and may drive the vasomotion underlying glymphatic clearance (Hauglund et al., 2025). Such questions depend on brief events, such as microarousals and state transitions, that span a few seconds. Scoring them means examining the same short interval across several signals and the animal's movement, and correcting boundaries across multi-hour recordings; without an integrated tool, this means moving among a signal viewer, predicted scores, and separate video software. One-second annotation is a requirement of this research, not a claim that boundaries or predictions are accurate to one second. The application is intended for experimenters scoring rodent EEG/EMG, particularly with aligned NE photometry and behavior video.

The current version reads a MATLAB file produced by a companion preprocessing pipeline, `preprocess_sleep_data`; other laboratories may need a thin adapter to this file layout.

State of the field

Several open tools support rodent sleep scoring. AccuSleep and its Python successor AccuSleePy (Barger et al., 2019) pair a manual-labeling interface with a neural-network scoring model. SPINDLE (Miladinović et al., 2019) provides end-to-end learned scoring across laboratories and species, now as a web service. Somnotate (Brodersen et al., 2024) scores with linear discriminant analysis and a hidden Markov model. Visbrain Sleep (Combrisson et al., 2019) is a general polysomnography viewer and hypnogram editor.

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These tools center on electrophysiology and, for most, on automatic scoring from EEG, EMG, or LFP features alone. None of their scoring methods uses NE photometry, although NE dynamics have been proposed as markers for distinguishing wake, microarousals, NREM, and REM sleep (Andersen et al., 2023). Beyond automatic scoring, our work needed a workspace for labeling and correcting scores, in which NE photometry and video checks inform scoring. Project 2 previously used Viewpoint's commercial SleepScore, which did not integrate NE photometry well; that gap motivated a dedicated application. It also had to suit many experimenters who are not programmers, yet every open tool above requires either setup in a programming environment or uploading recordings to a web service.

Software design

The application is designed around five tasks: (1) zero-setup installation and automatic updates, (2) visualizing and annotating at scale, (3) automatic scoring that keeps and learns from expert labels, (4) checking movement against video, and (5) saving, recovery, and export. Because smooth interactive visualization is central to these tasks, it is built with Dash and Plotly (Plotly Technologies Inc., 2015): a local web page, shown in a native window via pywebview and backed by a Python server on the same computer.

1. Zero-setup installation and automatic updates

We packaged the Windows application with PyInstaller so that it installs like conventional desktop software: users download a release, extract it, and double-click to open it. macOS users currently install from source, because a comparable package would need Apple's code signing and notarization. To avoid a rebuild and a fresh download for every change, the application's own source code stays outside the packaged runtime, and an integrated updater, released as the reusable, dependency-free package `desktop_app_source_updater` (Zhao, 2026a), fetches and applies bug fixes and new features at startup.[1]

2. Visualizing and annotating at scale

The EEG spectrogram (with a theta/delta ratio), EEG, EMG, and NE share one time axis, and the score is drawn as an overlay on every signal row (Figure 1A), so each signal can be used on its own to examine or label sleep states. Shortcut gestures minimize steps: for example, a right-click selects a whole contiguous bout, and a drag that reaches the edge of a zoomed-in view keeps panning, so a selection can extend arbitrarily far. A single key press then labels the selection.

Plotly alone becomes unresponsive when drawing the millions of samples in a multi-hour recording, so the application decimates long signals on demand with Plotly Resampler (Van Der Donckt et al., 2022): the server keeps the full-resolution signals and sends the interface only enough points to draw the visible range. Because each round trip to the server adds delay, selecting and labeling run entirely in the interface, and only zooming and panning request new data. Plotly Resampler's stock update path requests refreshed traces on every zoom or pan event, which compounds the delay during continuous movement; users of earlier versions reported lags of one to two seconds that disrupted scoring. We replaced that path: during continuous movement, the application waits for a pause before requesting new traces, drops requests already overtaken by newer movement, and swaps in only the changed trace data rather than redrawing the figure.

In manual timing tests on a 2.86-hour recording with v0.17.4, the median time from a navigation request to refreshed traces on a Windows laptop was 212 ms for mouse panning and 329 ms for keyboard navigation, down from about 935 ms in an earlier version (not a controlled

[1] Updates that change the packaged dependencies still require downloading a new release.

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benchmark). This interaction design is not specific to sleep; we extracted it into a runnable template for general multichannel time series (Zhao, 2026d).

3. Automatic scoring that keeps and learns from expert labels

The default scoring model is a rule-based algorithm that encodes the stage criteria used by Project 2 researchers (Kjaerby et al., 2022): it identifies Wake from EEG spectral power and REM from deep NE valleys, and without usable NE it identifies only Wake and NREM. Its thresholds adapt to each recording. Each run treats the user's labels (a single labeled interval is enough) as ground truth and chooses the settings that best fit them while changing the rest of the predictions as little as possible (Figure 1), so adaptation needs no multi-recording training set. Agreement with the user's labels shows only that the fit succeeded, not how accurate the predictions are elsewhere in the recording.

Alternatively, the deep-learning model sDREAMER (Chen et al., 2023), in a version adapted by the author (`sdreamer_flow`), runs when the optional PyTorch (Paszke et al., 2019) dependencies are installed.

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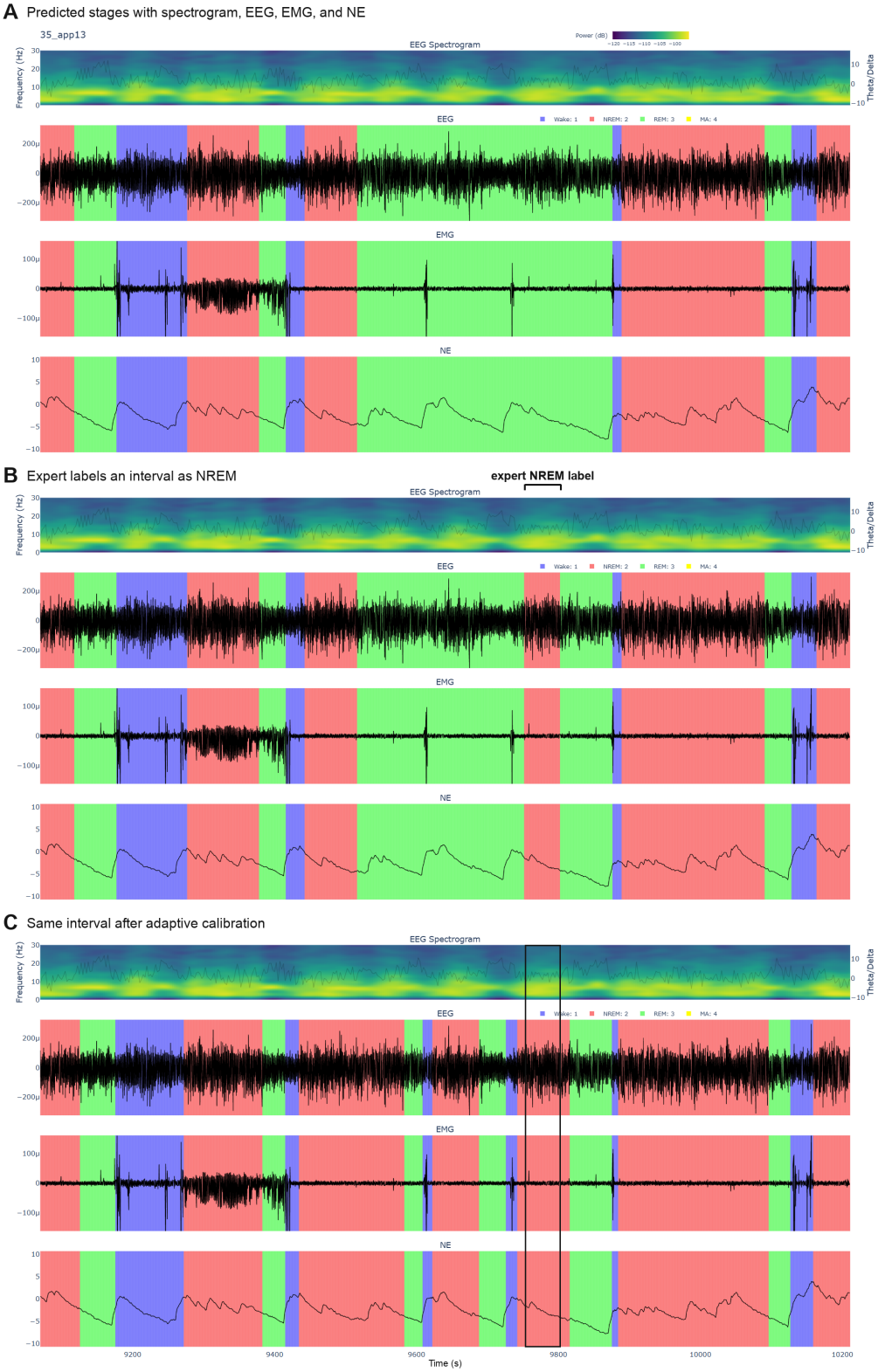


Figure 1: Expert-guided adaptation. The same interval at three successive steps: **(A)** the initial predicted stages over the spectrogram, EEG, EMG, and NE; **(B)** an expert relabels 50 s as NREM; **(C)** the model, recalibrated on 228 s of expert labels (178 s Wake outside this view, 50 s NREM), updates its predictions, including beyond the outlined interval.

4. Checking movement against video

Movements indicated by EMG help distinguish sleep states, and users sometimes corroborate them with behavior video. For a selected interval of up to five minutes, the application cuts the matching clip with FFmpeg and plays it in the same window (Figure 2).

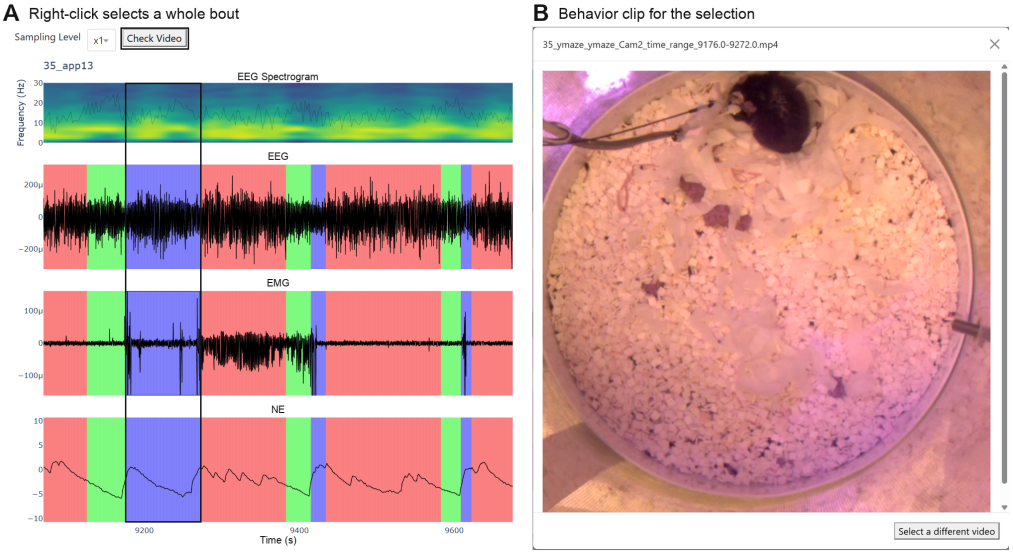


Figure 2: Selection-linked video review. (A) A right-click selects the bout at 9176–9272 s, and Check Video (outlined) opens its clip. **(B)** The 96-second clip plays beside the signals.

5. Saving, recovery, and export

Scores are saved back to the recording, and the most recent annotation or prediction can be undone. Saving a completely scored recording also exports bout, stage, and transition statistics to a spreadsheet; otherwise, saving reports what is left unscored, including short gaps hard to spot when zoomed out. After a crash, filesystem-backed recovery restores unsaved scores when the file is reopened.

Research impact statement

Researchers in Project 2 of the University of Rochester BrainFlowZZZ program have been the application's main users since 2023, using it to score their rodent EEG/EMG/NE recordings. The program's channel for the application has about 30 members; five users who reported their use have together scored about 1,200 hours, a lower bound on total use. The repository provides releases, documentation, tests, and demonstration videos, and links to archived versions (Zhao, 2026c) and an example dataset (Zhao, 2026b).

AI usage disclosure

From 2023 through 2025, the author asked web-based ChatGPT and Claude coding questions (model versions not recorded). Since 2026, coding agents have assisted with code, tests, documentation, and release checks: Claude Code (Claude Opus 4.5–5.5 and Fable 5) and OpenAI Codex (GPT-5 and GPT-6), with sessions recorded in the repository's work log (`treaty_docs/`), kept under the author's Agent Collab Treaty conventions for human–agent collaboration. Claude Code and Codex also helped plan, draft, and review this manuscript. The author made the architectural and design decisions, reviewed and tested all AI-assisted

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code before release, and reviewed, revised, and verified the manuscript against the software and cited sources.

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References

Andersen, M., Tsopanidou, A., Radovanovic, T., Compere, V., Hauglund, N., Nedergaard, M., & Kjaerby, C. (2023). Using fiber photometry in mice to estimate fluorescent biosensor levels during sleep. *Bio-Protocol*, *13*(15), e4734. https://doi.org/10.21769/BioProtoc.4734

Barger, Z., Frye, C. G., Liu, D., Dan, Y., & Bouchard, K. E. (2019). Robust, automated sleep scoring by a compact neural network with distributional shift correction. *PLOS ONE*, *14*(12), e0224642. https://doi.org/10.1371/journal.pone.0224642

Brodersen, P. J. N., Alfonsa, H., Krone, L. B., Blanco-Duque, C., Fisk, A. S., Flaherty, S. J., Guillaumin, M. C. C., Huang, Y.-G., Kahn, M. C., McKillop, L. E., Milinski, L., Taylor, L., Thomas, C. W., Yamagata, T., Foster, R. G., Vyazovskiy, V. V., & Akerman, C. J. (2024). Somnotate: A probabilistic sleep stage classifier for studying vigilance state transitions. *PLOS Computational Biology*, *20*(1), e1011793. https://doi.org/10.1371/journal.pcbi.1011793

Chen, J., Yao, Y., Anderson, M., Hauglund, N., Kjaerby, C., Untiet, V., Nedergaard, M., & Luo, J. (2023). sDREAMER: Self-distilled mixture-of-modality-experts transformer for automatic sleep staging. *2023 IEEE International Conference on Digital Health (ICDH)*, 131–142. https://doi.org/10.1109/ICDH60066.2023.00028

Combrisson, E., Vallat, R., O'Reilly, C., Jas, M., Pascarella, A., Saive, A.-L., Thiery, T., Meunier, D., Altukhov, D., Lajnef, T., Ruby, P., Guillot, A., & Jerbi, K. (2019). Visbrain: A multi-purpose GPU-accelerated open-source suite for multimodal brain data visualization. *Frontiers in Neuroinformatics*, *13*, 14. https://doi.org/10.3389/fninf.2019.00014

Hauglund, N. L., Andersen, M., Tokarska, K., Radovanovic, T., Kjaerby, C., Sørensen, F. L., Bojarowska, Z., Untiet, V., Ballestero, S. B., Kolmos, M. G., Weikop, P., Hirase, H., & Nedergaard, M. (2025). Norepinephrine-mediated slow vasomotion drives glymphatic clearance during sleep. *Cell*, *188*(3), 606–622.e17. https://doi.org/10.1016/j.cell.2024.11.027

Kjaerby, C., Andersen, M., Hauglund, N., Untiet, V., Dall, C., Sigurdsson, B., Ding, F., Feng, J., Li, Y., Weikop, P., Hirase, H., & Nedergaard, M. (2022). Memory-enhancing properties of sleep depend on the oscillatory amplitude of norepinephrine. *Nature Neuroscience*, *25*(8),

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1059–1070. https://doi.org/10.1038/s41593-022-01102-9

Miladinović, Đ., Muheim, C., Bauer, S., Spinnler, A., Noain, D., Bandarabadi, M., Gallusser, B., Krummenacher, G., Baumann, C. R., Adamantidis, A., & others. (2019). SPINDLE: End-to-end learning from EEG/EMG to extrapolate animal sleep scoring across experimental settings, labs and species. *PLOS Computational Biology*, *15*(4), e1006968. https://doi.org/10.1371/journal.pcbi.1006968

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., & others. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems 32*, 8024–8035. https://proceedings.neurips.cc/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html

Plotly Technologies Inc. (2015). *Collaborative data science*. Plotly Technologies Inc. https://plot.ly

Van Der Donckt, J., Van Der Donckt, J., Deprost, E., & Van Hoecke, S. (2022). plotly-resampler: Effective visual analytics for large time series. *2022 IEEE Visualization and Visual Analytics (VIS)*, 21–25. https://doi.org/10.1109/VIS54862.2022.00013

Zhao, Y. (2026a). *desktop_app_source_updater: A dependency-free source updater for packaged python desktop applications*. Zenodo. https://doi.org/10.5281/zenodo.21763329

Zhao, Y. (2026b). *Sleep Scoring App: Example EEG/EMG/NE recording and behavior video* [Data set]. Zenodo. https://doi.org/10.5281/zenodo.23489641

Zhao, Y. (2026c). *Sleep Scoring App: Interactive review and correction of rodent sleep annotations with synchronized electrophysiology, photometry, and behavior video*. Zenodo. https://doi.org/10.5281/zenodo.21748494

Zhao, Y. (2026d). *timeseries_app_cookbook: A runnable template and recipe book for building interactive desktop apps that navigate and annotate long time series*. Zenodo. https://doi.org/10.5281/zenodo.21763327

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